

Loop Mega Universe (LMU) — Pre-registered Falsifiers

A time-stamped registration of LMU’s falsifiable predictions, fixed before the deciding data.

- **Framework:** Loop Mega Universe (LMU) — a cyclic black-hole cosmology (a *lens/synthesis*, not a new predictive theory).
 - **Document of record:** `LMU_V3_28_consolidated.pdf` (V3.28, 2026-07-08). Where this registration and the `.tex` disagree, the `.tex` governs.
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 - **Registered (OSF):** doi.org/10.17605/OSF.IO/CN3B4 — the frozen, time-stamped pre-registration.
 - **Concept DOI:** [10.5281/zenodo.20692157](https://doi.org/10.5281/zenodo.20692157) · **Source:** github.com/wow116sk-collab/Loop-Mega-Universe · **License:** CC-BY-4.0
 - **Numbers reproduce from:** `code/verify_all.py` (CODATA spine, Klein–Gordon *m*-scan, relic statistics) — ALL PASS at V3.28.
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Why this document exists (the point of pre-registration)

The deciding measurements for LMU’s distinctive predictions are largely **still to come** — DESI-Y5 (~ 2027) and DESI DR3 / Euclid / Roman; X-ray-reflection / NewAthena + LISA (late 2030s); LiteBIRD-class B-mode (2030s); ongoing CMB anomaly-pattern searches. This file **fixes the thresholds now**, so a later “verdict right, numbers wrong” outcome cannot be papered over by moving the line after the fact (repository rule R6: a moved threshold is the worst failure mode). Each fixed threshold below is **immutable**: it must not be relaxed if data later push against the model.

Honesty preamble (recorded, not hidden). LMU’s *central* claim — that our aeon was seeded by the terminal flash of a previous aeon’s last black hole — is **directly unfalsifiable**: we cannot observe prior aeons. The lines below (F1, F2, F4, F5 live with fixed thresholds; F6 a watch-line; F3 retired) therefore test the framework’s **distinctive consequences within our own aeon**, not the cycle itself. **Note on numbering:** these F-labels follow the public README set; the document’s *internal* F-labels differ (e.g. its internal “F4” is the Hawking-point test = F5 here). Several lines were **reframed for the framework’s changed reading** (deterministic-flash reset; tiny-de-Sitter endgame) — see F5, F6, and the Registration statement. LMU contains **no new equation**; its one quantitative prediction (the tensor ratio r) is Starobinsky’s, inherited *conditional on the wiring* ($\omega =$ the prior aeon’s conformal factor = the α -attractor inflaton, status [Hypo $\rightarrow$ soft/favoured]).

Label legend: [Fact] measured/peer-reviewed · [Fact-th] from established theory · [Hypo] the assembly/wiring · [Open] unsolved.

F1 — Dark energy: canonical thawing quintessence, no phantom crossing

- **Prediction.** LMU’s dark-energy field A is **canonical quintessence with $w(z) \geq -1$ at all z and cannot cross -1** (Vikman 2005). A thawing field matches DESI’s w_0 **or** w_a , not both cleanly; DESI’s apparent $w < -1$ crossing is read as a **CPL-parametrisation artifact** (physical non-phantom thawing fits comparably — Dinda–Maartens 2025; de Souza et al. 2025). [Fact-th] / [Hypo] (the reading).

- **Fires (falsified) IFF:** DESI-Y5 (~ 2027) sustains $w(z) < -1$ at high z at $\gtrsim 3\sigma$, with systematics resolved, and the crossing survives a physical (non-CPL) reanalysis.
- **Does NOT fire if:** the data are $w \geq -1$ but evolving; a phantom hint is $< 3\sigma$ or systematics-limited; the crossing appears only under CPL.
- **Scope (what F1 kills):** a confirmed non-CPL crossing falsifies the **canonical** A -field (the spine’s choice); a quintom / k-essence upgrade would survive it, at the cost of a ghost. So F1 kills the *canonical* field, not every dark-energy variant.
- **The 3σ line is fixed now and must not be moved.**
- **Test / timeline:** DESI-Y5, ~ 2027 .
- **Status at registration (2026-07):** DESI DR2 rejects Λ CDM at $2.8\text{--}4.2\sigma$ with a CPL crossing near $z \sim 0.5$ — this **sharpens** F1 but does **not** fire it (CPL artifact; defence holds). One item to watch.

F2 — Black-hole spin: isolated over-massive relics spin *high*, merger-built BCGs spin *low*

- **Prediction.** The model forces **isolated over-massive relics (NGC 1277, Mrk 1216) to spin HIGH** (accretion-frozen, bucking the general high-mass $\rightarrow$ low-spin trend), while **merger-built cluster BCGs spin LOW** — a *decoupling* of spin from mass. Conditional on **coherent cocoon feeding** $f_{\text{coh}} \gtrsim 0.15$ (open joint J3, [Open]). For the carry-through variant, the inherited spin axis tilts past 45° after the survivor accretes ~ 0.77 of its mass in the new aeon. [Fact-th] / [Open] (the f_{coh} condition).
- **Fires (falsified) IF any of:** (a) a local over-massive **isolated relic is measured to spin LOW**; (b) a **clear major merger is found in such a relic**; (c) the **super-Eddington dense-cocoon wind is shown to shut off the hole’s own accretion ($M\text{--}\sigma$ self-regulation), not merely the host’s star formation** — LID-568 currently shows the *latter* (host SF only), which is what the mechanism needs; the *former* would remove the isolated-relic over-mass route.
- **Sharper cluster-side threshold:** on the merger/cluster (low-spin) side, only a confirmed **near-maximal spin** $a_\star \gtrsim 0.9$ would tension the merger-spin-down expectation ($a_\star \sim 0.55$).
- **Test / timeline:** X-ray reflection spectroscopy of accreting relics (now $\rightarrow$ 2030s); NewAthena + LISA (late 2030s) for the carry-through axis; BHEX-era spin history.
- **Status at registration (2026-07):** M87* spin is **contested** — Drew et al. 2025 ($a_\star \gtrsim 0.8$, Doppler beaming, a lower limit) vs Wong et al. 2025 (polarimetry, disfavours high spin) — logged as an **offsetting flag** ([Open --- contested]), not a falsification. The cluster spin-down expectation ($a_\star \sim 0.55$) vs $a_\star \gtrsim 0.8$ is a tension (factor 2.6–10.6 $\rightarrow$ an aligned-merger fraction $\geq 62\%$).

F3 — $P(k)$ re-initialisation — RETIRED / re-scoped (recorded for honesty)

- The earlier “primordial power-spectrum re-initialisation” falsifier was **ruled and re-scoped at V3.12**, and is now **closed-conditional** (horizon exile: every finite wall/patch scale exits the H_∞ horizon in finite N). It is **not a live falsifier line** at V3.28. Listed so the registry states plainly which lines are active and which were retired. [Fact-th, conditional].

F4 — Single-instanton origin signature: a primordial tensor ratio in the α -attractor plateau band

- **Prediction.** The $\sim 10^{16}$ GeV ignition (an α -attractor plateau) leaves a primordial **tensor-to-scalar ratio in the plateau band**: $r \approx 0.0037$ ($\alpha = 1$, Starobinsky, $N = 57$) up to $r \approx 0.009$ ($\alpha = 2.44$); under ACT DR6’s higher N (69–81), $r \approx 0.0018$ ($\alpha = 1$). **All below the current bound** $r < 0.036$ – 0.038 (BICEP/Keck). The (n_s, r) point sits **on the α -attractor locus**; the ignition B-mode is the **only** channel that reaches us (direct-detection GW is symmetry-suppressed and buried ~ 5 – 7 orders below the astrophysical background). [Fact-th, borrowed, conditional on the wiring].
- **Fires (falsified) IF:** LiteBIRD-class sensitivity measures r **inconsistent with the α -attractor plateau band** — e.g. a robust detection of r **well above the plateau ceiling**, or an (n_s, r) locus **incompatible with a single-field plateau**.
- **Does NOT fire if:** r is measured anywhere in the plateau band (0.0018–0.009) consistent with the (n_s, r) attractor line, or a null below current sensitivity.
- **Sharper thresholds (from the document):** a single quadratic A -field serving as *both* inflaton and dark energy gives $r = 8/N \approx 0.14$ — **already excluded** by BICEP/Keck ($r < 0.036$), which is *why* a separate plateau generator (Starobinsky / α -attractor) is required. A **reverse** line: a firm $r < 10^{-3}$ would favour the ekpyrotic (Ijjas–Steinhardt) rival over an LMU-type plateau. **Scope:** the $\Phi_{\text{gen}} \sim 10^{16}$ GeV ignition scale is **not an axiom**, so a differently-measured r *shifts the scale* rather than killing the loop — F4 is a **comparative / differential** signature, not a make-or-break falsifier.
- **Test / timeline:** LiteBIRD-class B-mode polarisation, 2030s.
- **Status at registration (2026-07):** consistent (predicted r -band under the bound); untested at the required sensitivity. Caveat: r is **not** an LMU-original number — it is Starobinsky’s, inherited conditional on the wiring; a confirmation supports the α -attractor ignition, it does not *uniquely* single out LMU.

F5 — No CCC-style Hawking points or concentric circles (a clean departure from bare CCC)

- **Prediction.** LMU predicts the **ABSENCE** of Penrose-CCC-style **Hawking points** and **concentric low-variance circles** in the CMB. Two mechanisms give the absence: **horizon exile** (every *other* island’s terminal flash sits outside our horizon) and — decisively under the current reading — the **deterministic-flash reset ignites inflation**, whose α -attractor plateau (57+ e-folds) **erases** any prior-aeon imprint. A clean discriminator from *bare* CCC, which predicts the circles. [Fact-th, conditional on the wiring + inflation].
- **Fires (falsified) IF:** a **confirmed detection** of CCC Hawking points / concentric circles. Per the document’s own stakes flag, a confirmed detection does not merely re-open a sub-item — it **breaks horizon exile itself**.
- **Reframed for the changed thinking (2026-07):** the document’s older caveat — “absence is *predicted* only on the B-primary *nucleation* joint, merely *permitted* on the B-alt *rescale* joint” — is a **residue of the superseded barrier framing**. The reset is now a **deterministic flash** $\rightarrow$ **inflation**, and inflation erases prior-aeon structure regardless of the old nucleation-vs-rescale fork, so the absence is predicted under the current reading (firmer, not fork-conditional).
- **Test / timeline:** CMB anomaly-pattern searches (Gurzadyan–Penrose 2010; An–Meissner–Nurowski–Penrose 2020; Meissner–Penrose 2025), ongoing.

- **Status at registration (2026-07): contested; consistent with absence.** Detection claims disagree — AMNP 2020 report 99.98% significance; Jow & Scott 2020 find $\sim 87\%$ ($\sim 1\sigma$) after ring-size marginalisation. No confirmed detection stands. (A confirmed Gpc-scale cross-aeon structure — the Giant Arc / Big Ring, itself contested — would fall in the same basket.)

F6 — The endgame is a *tiny positive* de Sitter, not a future AdS crunch

- **Prediction (current reading).** LMU’s chosen endgame is a **tiny nonzero de Sitter**: $V_{\min} = \rho_{\Lambda} = (2.3 \text{ meV})^4$, **anchored to the measured Λ** , so dark energy **asymptotes to a positive constant** ($H_{\infty} > 0$, $w \rightarrow -1$) — *not* to zero (the older “Minkowski / $A \rightarrow 0$ ” body prose, now **superseded**) and *not* to a negative floor (a future AdS transition $\rightarrow$ crunch). [**Fact-th, conditional**]; the endgame choice is [**Design**].
 - **Fires (falsified) IF:** DESI DR3 / Euclid / Roman establish a **future AdS transition** (dark-energy density crossing to negative, $V_{\min} < 0$) — contradicting the positive-de-Sitter asymptote the endgame commits to.
 - **Reframed for the changed thinking (2026-07):** the document’s older “ $V_{\min} = 0$ (Minkowski) vs $V_{\min} < 0$ (AdS)” discriminator is **superseded on the $V_{\min} = 0$ side** — the framework now commits to a *tiny positive* de Sitter. The live discriminator is therefore **positive-dS (LMU) vs future-AdS (rival)**.
 - **NOT the same as** the field-wide “**de Sitter stability** ($\Gamma > 0$ vs $\Gamma = 0$)” question, which LMU shares and does **not** list as an open item. F6 is the observational **sign of $V_{\min}$** (positive vs negative), not the decay-rate question.
 - **Honest status (leans rival): no fixed σ threshold is pre-registered** for this line, and current data **lean against** it — Gialamas et al. 2025 report a $\sim 93.8\%$ preference for a future AdS transition. Registered here as a **live tension to watch** (DESI DR3 / Euclid / Roman era), not a fixed-threshold falsifier; shared with other positive- Λ models, not unique to LMU.
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Corroborate vs falsify — at a glance

Line	Corroborated if	Falsified (fires) iff	Decider (era)
F1 dark energy	$w \geq -1$ evolving; crossing is CPL-only	$w < -1$ at high z , $\geq 3\sigma$, systematics resolved, non-CPL	DESI-Y5 (~ 2027)
F2 BH spin	isolated relics spin high, BCGs low	isolated over-massive relic spins low; a major merger in one; cocoon wind shuts off the hole’s own accretion; or $a_* \gtrsim 0.9$ cluster-side	X-ray reflection; NewA-thena+LISA (late 2030s)
F3 $P(k)$	— (retired)	— (re-scoped at V3.12, closed-conditional)	—
F4 tensor r	r in 0.0018–0.009 on the (n_s, r) locus	r outside the plateau band / off the attractor line (comparative, not make-or-break)	LiteBIRD-class (2030s)
F5 no Hawking pts	sustained null on CCC circles	a confirmed Hawking-point / circle detection (breaks horizon exile)	CMB anomaly searches (ongoing)
F6 endgame sign (watch)	DE asymptotes positive (tiny dS)	a confirmed future AdS transition ($V_{\min} < 0$); no fixed σ ; leans rival now	DESI DR3 / Euclid / Roman

Explicitly NOT falsifier lines (recorded for honesty)

These are in the document but are **not** registrable falsifiers — stated so the registry cannot be read as claiming more than it does:

- **Dark-matter nature / the 5:1 baryon:DM ratio — matched, not predicted.** LMU reminds DM as a thermal relic each aeon; the ratio and relic nature are *inputs*, and direct-detection bounds (LZ/XENONnT) enter only as an exclusion corridor the model must fit inside. **No DM-mass or cross-section prediction.**
- **The re-initialisation targets — $n_s, A_s, r, \Omega_k, f_{\text{NL}}$, isocurvature** — are **declared inputs** the generator must hit (Planck 2018 values), *not* predictions. (F4’s r is the α -attractor *consequence* of the plateau, a distinct use.)
- **CMB spectral distortions (μ, y)** — LMU predicts *no* distortion; satisfied by tens of orders (FIRAS). A consistency pass, not a discriminator.
- **PTA / NANOGrav stochastic GW background** — *supports* the SMBH-merger assumption; a consistency/cleanliness bound, not a unique falsifier.
- **CEAW (Cosmic Entropy Acoustic Waves)** — a within-aeon L2 **consistency re-check** (channel weights Γ_j [Open]), explicitly *not* a unique falsifier or a growth knob.
- **The “too-early / lump-seeded” seed-flux channel** — a bare “carry-through prediction” with **no observable or threshold** in the document; the doc itself cautions that high- z AGN over-massiveness can be a virial-mass artifact (up to 1 dex) and anchors the relic claim on *dynamical* masses. Not registrable as-is.

Registration statement

These predictions and thresholds are registered as of **2026-07-08 (V3.28)**, before the deciding data. The **fixed, immutable** thresholds — F1’s 3σ non-CPL line; F2’s low-spin / major-merger / own-accretion-shutoff breakers (and the $a_\star \gtrsim 0.9$ cluster line); F4’s plateau band (with the $r = 0.14$ exclusion and the $r < 10^{-3}$ reverse line); F5’s *confirmed-detection* breaker — will **not** be relaxed if later data push against the model. F6 is registered as a **watch-line without a fixed σ threshold** (and currently leans rival), and F3 is stated as **retired** (re-scoped at V3.12), not silently dropped.

Reframed for the changed framework (important). The document changed its reset/endgame reading after the earlier body text was written (append-only): the reset is now a **deterministic flash** (not a barrier/nucleation-vs-rescale fork), and the endgame is a **tiny positive de Sitter** (not $V_{\min} = 0$ / Minkowski). The `.tex` body still carries the superseded prose *verbatim* (append-only), with the 3.27 revision entries as the reconcile. **F5 and F6 above are written against the *current* reading**, not the superseded body prose — which is why they had to be reframed (F5 firmer; F6 a positive-dS-vs-AdS sign test). SSOT for the current state: `review/LMU_SYNTHESIS_2026-07-07.md`.

The framework’s own immediate open item is **one linked initial-condition question** (the wiring, [Hypo $\rightarrow$ **soft/favoured**], carrying the relocated residue of the former “+3 spread”, now the standard flatness problem). Field-wide questions LMU shares — **de Sitter stability** (Γ), the measure problem, the swampland — are **not** LMU closure-blockers and are **not** falsifier lines (distinct from F6, the observational *sign* of $V_{\min}$).

Provenance (document of record, V3.28): §“Falsifiers” (F2 relic-spin); §“DESI status and the pre-registered falsifier” (F1, *verbatim*); the ignition-GW revision record (F4); Part VII §7 / ledger item (D) + Part IV B-primary (F5); Part VI §18 “Open joint 2 — sign of $V_{\min}$ ” + the 3.27 tiny-de-Sitter revision entries (F6). SSOT: `review/LMU_SYNTHESIS_2026-07-07.md`.